Anticipatory Scheduling Before Policy Learning: A Scoped Timing and Recovery Study

Research draft [1]

Abstract

Exact response identification previously corrected admission errors without improving physical timing. We now test an early, public eight-pair fit and anticipatory reference prediction in a two-process mechanical fixture. Across 360 new trajectories and 180 retained comparisons, early robust recovery reaches 51/180 joint timing passes versus 37/180 previously, with 14 paired repairs and no harms. Complete search of a declared, privileged expanded command graph permits 66/180 cases. The bound relaxes feedback guards and is not an online policy optimum. All new trajectories and 110 offline native witnesses are independently reconstructed. These results motivate stronger control and information baselines before claims about learned policies or chemical outcomes.

1. Predict early, measure independently

For each required process, actual reference τ_i and effect T_i must satisfy $T_i - \tau_i \in [L_i, U_i]$. Process B also requires continuity before its minimum duration. A selective channel serializes device commands; a reliable relay stops both processes. ACK does not establish physical completion. The original attempt verdict cannot be repaired retrospectively.

Only delivered velocity, power, timestamps and public service parameters enter the controller. The same noiseless implicit-Euler model and OLS support checks now freeze after eight powered pairs, rather than waiting for the reference message. A frozen snapshot predicts the reference interval under continued drive. The first A command may be reserved before that message arrives; its expected ACK preserves the predicted physical event. Measured references later supersede predictions in planning. Only actual persisted samples determine scoring.

We retain the complete-branch and full-supported-path-first rules. The latter admits a nominally supported retry only when no fully supported candidate exists. Both preserve B on observed failure. This is classical identification and deterministic control, with no new weight training.

Table 1. Joint physical timing over 180 cases per control.

Control	TRUE	FALSE	UNKNOWN
Prior robust recovery	37	141	2
Early complete branch	46	134	0
Early robust priority	51	129	0

2. Execution and paired attribution

Five response settings, three delivery delays, two A windows, two B windows and three faults define 180 cases per control. Two new controls execute 360 trajectories, recording 360,360 samples and 3,960 RGB frames. The actor does not read RGB. Explicit delays advance physics; host work is logged without advancing the simulated process, so no real-time claim is made.

An initial interface error occurs before the first completed case. Its failed release is preserved; the public initial service duration is bound in a separate corrected version and six checks precede the new freeze. The independent auditor later receives a path-resolution-only correction; control, samples and scoring remain frozen.

Early complete-branch control repairs 13 cases but harms four relative to the retained robust-priority comparator: branch priority also differs. An explicitly post-hoc matched comparison to the earlier complete-branch control gives 33 to 46 passes, with 13 repairs and no harms. Early robust priority repairs 14 and harms none against its matched retained priority. All controls maintain B continuity in all 180 cases.

3. A finite, privileged command bound

We define reference-reactive and early-fit request gates on a 10 ms grid. Both graphs preserve full-drive reference events, native service durations and FIFO, but use exact fault identity and relax failure-observation waits. At most

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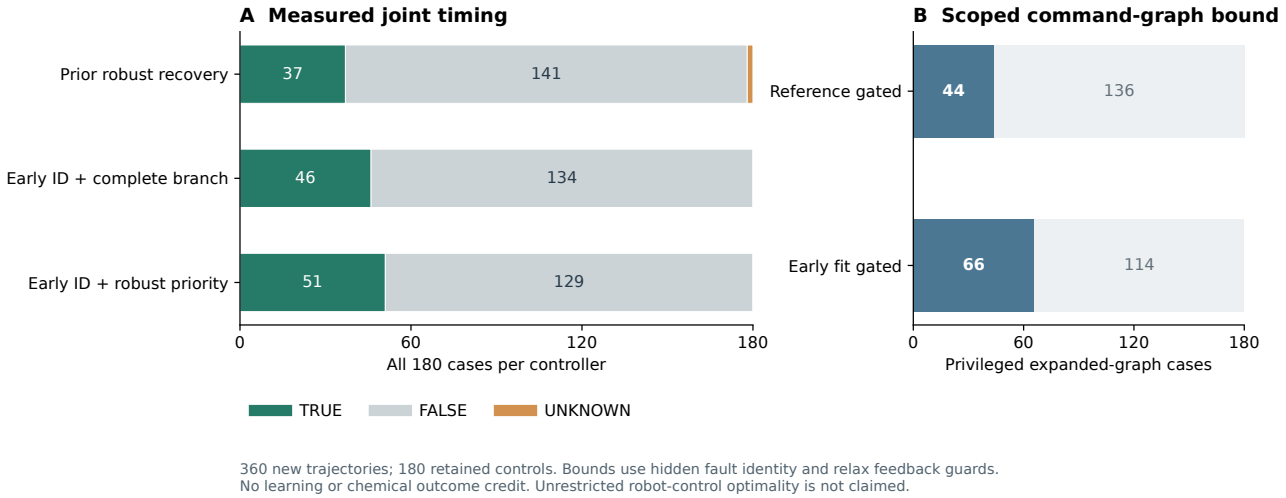


Figure 1. All cases remain in each denominator. The left panel reports independently measured TRUE/FALSE/UNKNOWN; the prior control has two UNKNOWN cases. The right panel concerns separately named privileged graphs, not causal execution success. The configurations are repeated deterministic development cases, not independent chemistry batches.

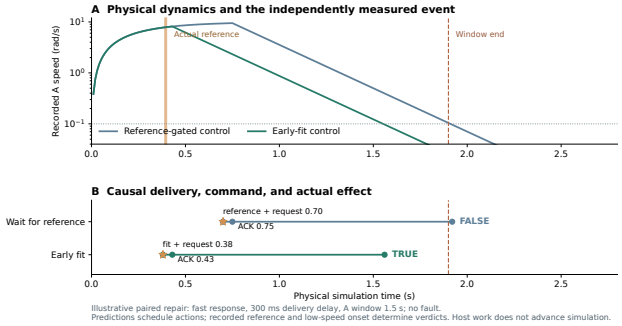


Figure 2. A paired repair under 300 ms delivery delay. Public fit delivery permits an earlier command; actual reference and effect intervals remain the independent scoring endpoints. This is a control improvement.

one initial A, one ordered retry, one B and one relay command are allowed. First physical zero times (a, b) suffice because shutdown is irreversible. Any later shutdown can be replaced by a relay without altering an already stopped process. This yields three normal forms: selective A then relay, selective B then relay, or one relay.

All grid pairs are exactly pruned by per-process timing and continuity; normal-form service and request gates are then exhaustively checked. A tiny command-history enumeration validates this reduction. The reactive expanded graph admits 44 cases, the early-fit expanded graph 66. All 540 main comparison trajectories lie in their corresponding graphs; 110 feasible witnesses are independently re-integrated with native commands. The 114 early-graph infeasible cases cannot pass within this reference- preserving scope. Feasibility of a privileged witness does not prove attainability with causal information.

4. What remains open

All new controller histories replay exactly; native reconstruction error is zero, response-fit error is below 2.4×10^{-16} and predicted reference midpoints match measured ones in this known noiseless family. The 15-case gap from early robust control to the privileged graph includes fault knowledge and relaxed feedback costs, and is not established learning headroom. Censored relay takeovers cannot label primary actuator success; public fault support must respect feedback arrival times. No policy learning, independent chemical endpoint, real robot or new task topology is demonstrated. Shared history-conditioned imitation and student-state relabeling remain future method experiments.